Semantic Segmentation of Remote Sensing Images With Inconsistent Resolutions via a Spectral-Geometric Iterative Fusion Network

Wenqi Han, Wen Jiang, *Member, IEEE*, Jie Geng, *Member, IEEE*, and Yanchen Bao

***Abstract*—The fusion of optical, hyperspectral, and synthetic aperture radar (SAR) images is essential for semantic segmentation in remote sensing, enabling more comprehensive land cover classification through multimodal data integration. However, disparities in spatial resolution and imaging characteristics across modalities impede effective feature alignment and fusion, degrading segmentation performance. To address this problem, we propose a novel spectral-geometric iterative fusion network (SGIFN), specifically designed to handle multimodal semantic segmentation with inconsistent resolutions. The core innovation of SGIFN lies in its unified architecture that progressively aligns, integrates, and enhances multimodal features through three newly designed modules. The spectral-spatial iterative decoupling (SSID) module introduces a novel iterative mechanism to adaptively align and decouple optical and hyperspectral features. The spectral-geometric synergistic conditional random field (SGS-CRF) module captures both local and long-range spatial dependencies by synergizing geometric (SAR) and spectral information. The class-guided multiscale contrastive aggregation (CG-MCA) module further strengthens feature representation across scales via multiclass, contrastive learning. We constructed a new multimodal remote sensing dataset comprising optical, hyperspectral, and SAR images with varying resolutions collected from Wuhan and Suzhou regions. Experimental results show that SGIFN achieves a mean Intersection over Union (mIoU) of 69.61% on Suzhou dataset and 63.96% on Wuhan dataset. These results demonstrate the effectiveness of SGIFN in handling multimodal data with inconsistent resolutions.**

***Index Terms*—Inconsistent resolutions, multimodal fusion, remote sensing, semantic segmentation.**

I. Introduction

REMOTE sensing image processing plays a crucial role in various applications, including geographic information systems (GISs) [1], [2], [3], [4], disaster management, and environmental monitoring [5]. The integration of advanced satellite and aerial sensor technologies has led to the proliferation of multimodal remote sensing data, encompassing optical [6], synthetic aperture radar (SAR) [7], hyperspectral [8], and thermal imagery [9]. These modalities provide complementary perspectives of the Earth's surface. Leveraging these diverse data sources facilitates comprehensive land cover analysis, infrastructure monitoring, and environmental assessments, significantly enhancing their applications in urban planning, agriculture, and disaster response.

Semantic segmentation plays a key role in remote sensing by assigning each pixel to a semantic category [10], [11]. It enables detailed land cover classification, such as identifying roads, buildings, and water bodies in urban areas. With the rise of multimodal data, combining optical, hyperspectral, and SAR imagery has become a promising approach to improve segmentation accuracy. However, the inherent differences in spatial resolution and imaging mechanisms across modalities, resulting from distinct sensor characteristics, present a significant challenge for effective multimodal fusion. Optical images usually provide high spatial resolution but lack spectral richness [12]; hyperspectral images, by contrast, offer detailed spectral signatures essential for material identification but often suffer from limited spatial resolution [13], [14]. SAR imagery contributes robust structural and textural features under all-weather conditions [15], yet typically at a lower resolution than optical sensors. These resolution mismatches lead to two major technical issues: feature misalignment and fusion complexity. Misalignment can distort spatial or spectral structures and compromise segmentation accuracy [16], [17], while the fusion process itself is further complicated by the heterogeneous nature of the data—spatial geometry versus spectral signature [18]. To mitigate these challenges, a growing body of research has focused on more sophisticated fusion mechanisms. Rather than relying on naive concatenation or uniform processing, recent methods explicitly model modality heterogeneity. For instance, Wei et al. [19] propose guiding multilevel semantic aggregation using structural priors or deep feature cues. Wang et al. [20] design asymmetric architectures that extract complementary features through modality-specific pathways and align them via transformer-based attention modules. Furthermore, Song et al. [21] introduce modified attention mechanisms to filter noise and refine cross-modal interactions, enhancing the quality of fused representations. These approaches share a common goal: improving fusion effectiveness by accounting for the unequal contributions of each modality.

While existing multimodal segmentation methods have advanced in fusion strategies, their performance remains limited by the quality and characteristics of available datasets.

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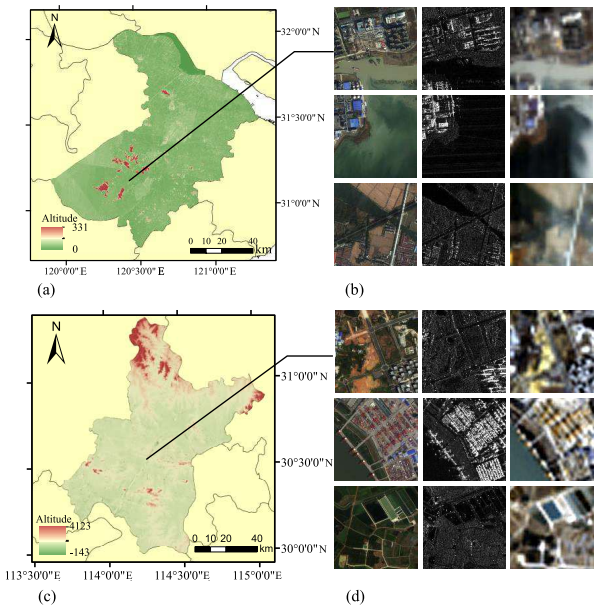


Fig. 1. Geographical locations of Suzhou and Wuhan and examples of multimodal data. (a) Suzhou's geographical location. (b) Sample image of Suzhou dataset. (c) Wuhan's geographical location. (d) Sample image of Wuhan dataset.

In particular, many widely used datasets—such as Houston 2018 [22] and the cross-domain dataset by Hong et al. [23]—uniformly resample all modalities to the same resolution. Although this simplifies data processing and model training, it sacrifices the native resolution characteristics of each modality, hindering models from learning resolution-aware representations. To address the limitations of uniform-resolution datasets, we construct a new multimodal remote sensing dataset using imagery from Gaofen-2, Gaofen-3, and Gaofen-5 satellites, covering regions in Wuhan and Suzhou. Fig. 1 shows the geographical distribution and representative examples of the data. Unlike existing datasets that resample all modalities to a common scale, our dataset preserves the native resolutions of optical, SAR, and hyperspectral images while ensuring accurate spatial alignment. This design enables more realistic evaluation of multimodal fusion under resolution discrepancies and provides a valuable benchmark for resolution-aware representation learning.

Our primary motivation is to address the inconsistencies in spatial detail and spectral representation caused by resolution gaps and modality disparities in multimodal semantic segmentation. To this end, we propose the spectral-geometric iterative fusion network (SGIFN) for multimodal remote sensing imagery with inconsistent resolutions. Based on this objective, our article makes the following key contributions: First, to alleviate feature misalignment resulting from spatial resolution differences between hyperspectral and optical images, we introduce a spectral-spatial iterative decoupling (SSID) module, which progressively aligns and integrates spectral-spatial features through 3-D convolution and grouped correlation. Second, to effectively fuse the geometric information from SAR with spectral features while capturing long-range dependencies, we propose a spectral-geometric synergistic conditional random field (SGS-CRF) module. It employs fully connected conditional random fields (CRFs) with a window-based optimization strategy to balance spatial detail preservation and computational efficiency. Third, to address inconsistencies in cross-scale representations, we design a class-guided multiscale contrastive aggregation (CG-MCA) module that enhances intra-class compactness and inter-class separability through multiclass contrastive learning. Finally, recognizing the lack of benchmark datasets that maintain native resolution disparities across modalities, we construct a new multimodal remote sensing dataset covering Wuhan and Suzhou regions. This dataset includes co-registered optical, SAR, and hyperspectral images with their inherent resolution differences preserved. The overview of SGIFN is shown in Fig. 2.

The structure of this article is organized as follows: Section II reviews relevant literature on multimodal semantic segmentation. Section III outlines the proposed semantic segmentation framework, elaborating on its architectural design and key components. Section IV presents the experimental results on the datasets, including a comparative analysis of the proposed method against existing approaches. Finally, Section V concludes the article by summarizing the findings and proposing potential directions for future research.

II. Related Work

A. Semantic Segmentation

Semantic segmentation is a critical task in computer vision, aiming to assign a semantic category label to each pixel in an image [24]. Over the years, significant progress has been made in semantic segmentation methods, driven by advancements in deep learning techniques. UNet [25] is a milestone in this field, introducing a symmetric encoder-decoder architecture with skip connections to achieve high-precision pixel-level predictions, making it a benchmark model for tasks such as medical image segmentation. Following UNet, improvements based on deep convolutional networks further advanced the field. For instance, the DeepLab series [26] expanded the receptive field using atrous convolutions and atrous spatial pyramid pooling (ASPP), while HRNet [27] maintained high-resolution feature maps and fused multiscale features in parallel, significantly enhancing performance in complex scenes. More recently, the introduction of Transformers has brought new breakthroughs to semantic segmentation. SegFormer [28] achieved efficient segmentation performance with a lightweight Transformer encoder and multilayer perceptron (MLP) decoder. The latest VMamba [29], based on the state space model (SSM), dynamically selects important features through a selective scan mechanism, achieving a better balance between long-range dependency modeling and computational efficiency, thus emerging as a new paradigm in semantic segmentation.

In the field of remote sensing, semantic segmentation has demonstrated significant potential for applications such as land-use classification, building extraction, and vegetation cover monitoring in high-resolution imagery. A particularly promising direction is the advancement of multimodal semantic segmentation, which seeks to integrate complementary

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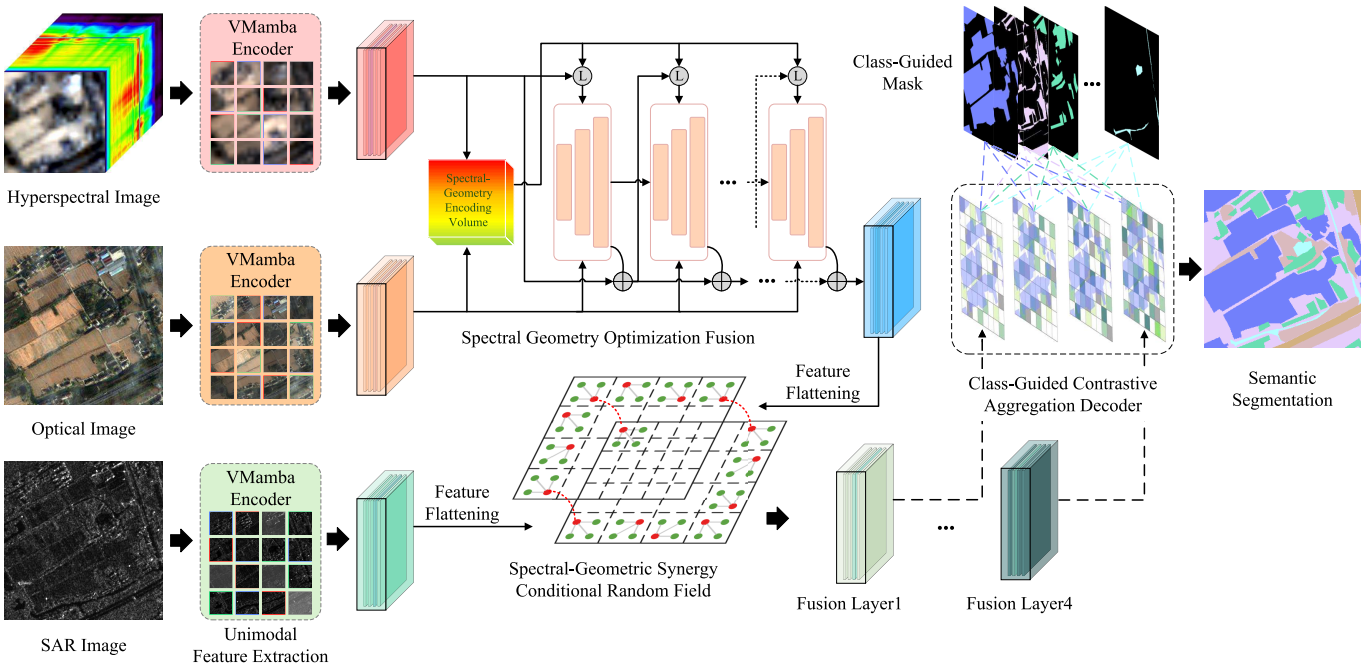


Fig. 2. Our framework of SGIFN for semantic segmentation of inconsistent resolutions remote sensing images.

information from diverse data sources, including optical, radar, and LiDAR data. However, a critical challenge in multimodal segmentation lies in the effective fusion of heterogeneous features from different modalities. Overcoming these limitations could unlock new possibilities for more accurate and robust semantic segmentation in complex remote sensing scenarios.

B. Multimodal Semantic Segmentation for Remote Sensing

Recent advancements in multimodal semantic segmentation have explored various strategies to address the challenges associated with integrating heterogeneous data sources. These methods generally fall into two categories: feature-level fusion [30], [31], [32], decision-level fusion [33]. Feature-level fusion is one of the most widely adopted approaches, where data from different modalities are first processed to extract features, which are then integrated for segmentation. For instance, Hong et al. [34] introduced a convolutional neural network (CNN)-based fusion framework that combines spatial features from high-resolution optical imagery with spectral features from hyperspectral data to enhance classification performance. Similarly, Li and Shen [35] proposed a dual-branch network to independently process optical and SAR data, followed by a cross-attention mechanism to align and fuse features from both modalities effectively. Liu et al. [36] introduce SoftFormer, a novel network that combines the strengths of CNNs and transformers, incorporating interior self-attention for local feature extraction and a joint key–value learning fusion for multilevel integration of multimodal data. Decision-level fusion methods, in contrast, combine predictions from separate segmentation models trained on individual modalities. For example, Yu et al. [37] developed a cross-domain ensemble learning approach that aggregates segmentation results from RGB and deep modalities using a probabilistic voting mechanism. While simpler to implement, decision-level fusion often struggles to fully exploit the complementary characteristics of different modalities. Cross-modal learning has also gained attention, aiming to leverage one modality to enhance another through tasks such as super-resolution or feature augmentation. For instance, Gao et al. [38] proposed a hyperspectral super-resolution framework that utilizes high-resolution optical images to guide the spatial enhancement of hyperspectral data. This method employs a joint loss function to maintain spectral fidelity while improving spatial resolution, thus facilitating more accurate segmentation. Another notable approach is the work by Wang et al. [39], which employs a shared feature space to align optical and SAR data, enabling robust segmentation even under incomplete or noisy input conditions.

Despite significant advancements in multimodal semantic segmentation of remote sensing images, challenges stemming from resolution discrepancies between different modalities have yet to be fully addressed. Therefore, there is an urgent need for the development of semantic segmentation models that can effectively handle resolution mismatches, which is essential for enhancing the performance and robustness of multimodal segmentation in remote sensing applications.

III. Methodology

A. Problem Formulation

In the task of multimodal semantic segmentation with inconsistent resolutions, suppose we have a series of labeled training data $D_l^0, D_l^1, \ldots, D_l^n$, where $D_l^i = \{I_{\text{opt}}^i, I_{\text{sar}}^i, I_{\text{hsi}}^i, y_i\} (0 \leq i \leq n)$, I_{opt}^i represents the optical image, I_{sar}^i represents the SAR image, I_{hsi}^i represents the hyperspectral image, and y_i is the corresponding semantic segmentation ground truth image. I_{opt}^i and I_{sar}^i have the same resolution and identical

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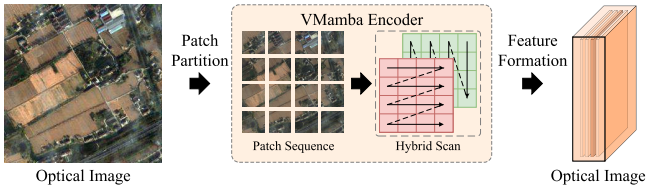


Fig. 3. Overview of the VMamba-based feature extraction.

input dimensions, while I^i_{hsi} differs in both resolution and input dimensions compared to I^i_{opt} and I^i_{sar}. Our objective is to perform multimodal semantic segmentation by leveraging images from different modalities and resolutions.

We propose an SGIFN for multimodal remote sensing image semantic segmentation under resolution discrepancies. We train the network on multimodal data with resolution inconsistencies, denoted as $D^0_l, D^1_l, \ldots, D^n_l$. First, for each training batch, optical and hyperspectral images are fed into the SSID module. This module decouples and integrates spectral features across modalities using grouped correlation volumes, geometric encoding, and a lightweight 3-D UNet (Section III-B). Next, the fused spectral features and SAR image features are input into the SGS-CRF module, which effectively integrates SAR spatial geometry with spectral features (Section III-C). Finally, we introduce the CG-MCA module to refine the predictions, completing the semantic segmentation task under resolution discrepancies (Section III-D).

B. Spectral-Spatial Iterative Decoupling

In the process of multimodal feature fusion between optical and hyperspectral imagery, the high spatial resolution of optical images can significantly enhance the spatial details of hyperspectral images. Meanwhile, the rich spectral information provided by hyperspectral images aids in distinguishing materials and object categories. However, the disparity in resolution between these two modalities presents substantial challenges for effective cross-modal feature fusion. To address these challenges, this article proposes a novel SSID module that constructs a spectral-spatial encoding volume. This module iteratively optimizes the decoupling of hyperspectral image features using optical imagery to resolve the misalignment of features caused by resolution discrepancies between optical and hyperspectral data.

First, initial feature extraction is conducted on both optical and hyperspectral images to obtain foundational representations. To this end, we employ the VMamba encoder f_{VMamba}, a recently introduced architecture based on the Selective SSM [29]. In contrast to traditional Transformer-based encoders that incur quadratic computational complexity due to self-attention mechanisms, VMamba incorporates a selective scanning strategy that compresses input features while retaining salient semantic information. This enables efficient modeling of long-range dependencies with reduced computational overhead. The VMamba encoder demonstrates strong adaptability to multimodal data, as it can process both three-channel optical images and high-dimensional hyperspectral images (e.g., with 330 bands) in a unified manner. In our framework, a shared VMamba backbone is utilized to extract hierarchical feature representations from both modalities. This backbone outputs features at four progressive levels of abstraction, facilitating the capture of both fine-grained local patterns and broader contextual information. Such hierarchical representations serve as a critical foundation for subsequent cross-modal fusion and semantic segmentation. An overview of the VMamba-based feature extraction process is illustrated in Fig. 3, which highlights the selective scanning mechanism and the nature of the extracted features. The feature extraction process can be mathematically represented as follows:

$$\mathbf{F}_{\text{opt}} = f_{\text{VMamba}}\left(\mathbf{I}_{\text{opt}}\right) = \left\{F^1_{\text{opt}}, F^2_{\text{opt}}, F^3_{\text{opt}}, F^4_{\text{opt}}\right\} \tag{1}$$

$$\mathbf{F}_{\text{hsi}} = f_{\text{VMamba}}\left(\mathbf{I}_{\text{hsi}}\right) = \left\{F^1_{\text{hsi}}, F^2_{\text{hsi}}, F^3_{\text{hsi}}, F^4_{\text{hsi}}\right\} \tag{2}$$

where $\mathbf{I}_{\text{opt}}$ and $\mathbf{I}_{\text{hsi}}$ denote the input optical and hyperspectral images, respectively, and F^i_{opt} and F^i_{hsi} represent the feature maps obtained at different levels of the VMamba encoder.

In the process of decoupling hyperspectral features with the assistance of optical images, we construct a spectral-spatial encoding volume to effectively capture inter-modal feature relationships. Taking F^1_{opt} and F^1_{hsi}, as examples, we construct grouped correlation volume. We divide the features F^1_{opt} and F^1_{hsi} into N_g groups (assuming $N_g = 8$) along the channel dimension and calculate the correlation maps group by group. We define the feature correlation volume $\mathbf{C}_{\text{corr}}$ as

$$\mathbf{C}_{\text{corr}}(g,k,x,y) = \frac{1}{N_c/N_g}\left\langle \mathbf{F}^1_{\text{opt}}(g,x,y), \mathbf{F}^1_{\text{hs}}(g-k,y)\right\rangle \tag{3}$$

where $\langle\cdot,\cdot\rangle$ denotes the inner product operation, k represents the feature matching index, and N_c is the total number of feature channels. While this feature correlation volume captures local correlations, it does not fully account for the global geometric structure. To address this limitation, we apply a lightweight regularization network R to process $\mathbf{C}_{\text{corr}}$, yielding the geometry encoding volume $\mathbf{C}_G$

$$\mathbf{C}_G = \mathbf{R}\left(\mathbf{C}_{\text{corr}}\right). \tag{4}$$

The regularization network R is built upon a lightweight 3-D UNet architecture, consisting of three down-sampling blocks and three up-sampling blocks. Each down-sampling block includes two $3 \times 3 \times 3$ 3-D convolutions, with the number of channels increasing from 16 to 48 across the three blocks. Each up-sampling block features a $4 \times 4 \times 4$ 3-D transposed convolution followed by two $3 \times 3 \times 3$ 3-D convolutions. We employ the CoEx method [40] to excite the cost volume channels using optical features for guided aggregation. For a given cost volume $\mathbf{C}_i$ of size $D_i \times H_i \times W_i$ (where $i = 4, 8, 16$, and 32), the guided excitation is expressed as

$$\mathbf{C}'_i = \sigma\left(\mathbf{F}^i_{\text{opt}}\right) \odot \mathbf{C}_i \tag{5}$$

where σ denotes the sigmoid function, and $\odot$ represents the Hadamard product. This guided excitation enables the regularization network to propagate scene geometry information, resulting in a refined geometry encoding volume. Next, we compute all-pairs correlations between optical and hyperspectral features to capture local feature relationships. To expand the receptive field, we apply 1-D average pooling along the

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disparity dimension, with a kernel size of 2 and a stride of 2. This pooling operation forms a multilevel pyramid of geometry encoding volumes and correlation volumes, which are then combined to produce a final, enriched geometry encoding volume.

To initiate cross-modal decoupling, we first estimate an initial alignment index $\mathbf{d}_0$ from the spectral-spatial encoding volume $\mathbf{C}_G$ using a soft argmin operation

$$\mathbf{d}_0 = \sum_{d=0}^{D-1} d \times \text{Softmax}\left(\mathbf{C}_G\left(d\right)\right) \tag{6}$$

where d denotes the predetermined alignment indices corresponding to spatial positions at a reduced resolution. Following this, we employ a multilevel ConvGRU framework, consisting of three ConvGRU layers, to iteratively refine the alignment prediction. Each ConvGRU layer is initialized with multiscale context features from the spectral-spatial encoding volume.

At each iteration k, the current alignment index $\mathbf{d}_k$ is used to extract a set of geometry features $\mathbf{G}_f$ by linearly interpolating values from both the geometry encoding volume $\mathbf{C}_G$ and the correlation volume $\mathbf{C}_A$. This process enables us to capture comprehensive spatial and spectral information. The geometry features are computed as follows:

$$\mathbf{G}_f = \sum_{i=-r}^{r} \text{Concat}\left\{\mathbf{C}_G\left(\mathbf{d}_k + i\right), \mathbf{C}_A\left(\mathbf{d}_k + i\right) \right. \\ \left. \mathbf{C}_G^p\left(\mathbf{d}_k/2 + i\right), \mathbf{C}_A^p\left(\mathbf{d}_k/2 + i\right)\right\} \tag{7}$$

where r is the indexing radius, and p represents pooling. The geometry features $\mathbf{G}_f$ are then passed through the encoder layers to generate updated features $\mathbf{d}_k$, which are used to update the ConvGRU hidden state. Using the previous hidden state $\mathbf{d}_k$, a residual alignment adjustment $\Delta\mathbf{d}_{k+1}$ is predicted

$$\mathbf{d}_{k+1} = \mathbf{d}_k + \Delta\mathbf{d}_k. \tag{8}$$

At each iteration, the ConvGRU updates the decoupled feature representations by integrating both optical and hyperspectral image features, progressively enhancing the precision of cross-modal decoupling. This iterative process gradually combines the high spatial resolution of the optical images with the rich spectral information of the hyperspectral images, refining the joint feature representation of the decoupled output.

C. Spectral-Geometric Synergy CRF

Due to the fundamental differences in the information provided by SAR and spectral features, SAR images excel at capturing precise spatial geometry and texture, offering fine-grained spatial resolution, particularly in complex landscapes. In contrast, spectral data (from optical and hyperspectral imagery) provides detailed spectral signatures that effectively distinguish materials and land features, but often lacks the fine spatial resolution that SAR offers. Integrating these modalities requires not only aligning spatial resolution but also preserving their complementary nature—capturing detailed spatial structures from SAR while fully exploiting the spectral diversity of optical and hyperspectral data. To address this, we propose a fusion framework based on SGS-CRF. The framework takes spectral features obtained from the fusion of optical and hyperspectral image features as one input and spatially detailed features from SAR images as the other input. The SGS-CRFs framework comprises three key components: (a) energy modeling using fully connected CRFs (FC-CRFs) to capture local and long-range cross-modal dependencies; (b) windowed CRF optimization to reduce computational complexity by partitioning feature maps; and (c) iterative feature map updates through multiwindow aggregation to ensure feature consistency and model convergence.

Effectively fusing SAR and spectral features demands capturing both local similarities and long-range dependencies across modalities. We address this challenge by employing FC-CRFs to model relationships between features from different modalities, enhancing the fusion's effectiveness. This approach strengthens spatial and spectral consistency in regions with high alignment, improving accuracy in complex scenarios. The process begins by extracting geometric information from SAR images, which is integrated with spectral features using the FC-CRF model. To ensure effective feature extraction from the SAR images ($\mathbf{I}_{\text{sar}}$), we utilize the VMamba encoder to extract geometric information, represented as a hierarchical set of feature maps

$$\mathbf{F}_{\text{sar}} = f_{\text{VMamba}}\left(\mathbf{I}_{\text{sar}}\right) = \left\{F_{\text{sar}}^1, F_{\text{sar}}^2, F_{\text{sar}}^3, F_{\text{sar}}^4\right\}. \tag{9}$$

The energy function of the CRF balances unary potentials, representing the confidence of individual features, and pairwise potentials, enforcing spatial and spectral coherence. The energy function is defined as

$$E\left(X\right) = \sum_i \psi_u\left(x_i\right) + \sum_{i,j} \psi_p\left(x_i, x_j\right) \tag{10}$$

where $\psi_u(x_i)$ is the unary potential computed using a neural network that processes SAR and spectral image features, while $\psi_p(x_i, x_j)$ is the pairwise potential capturing the correlation between nodes i and j. The unary potential is expressed as

$$\psi_u\left(x_i\right) = \theta_u\left(F_{\text{sar}}^1, F_{\text{opt_hsi}}^1, x_i\right) \tag{11}$$

where θ_u is the learned neural network function that computes the unary potential. The function simultaneously considers SAR features as well as the spectral features of the fusion of optical and HSI (represented as $F_{\text{sar}}^1, F_{\text{opt_hsi}}^1$) and the current pixel x_i. By using neural networks, the unary potential can be flexibly learned from the high-dimensional feature maps of both SAR and HSI, enabling it to capture complex relationships between the two modalities' features. The purpose of the pairwise potential $\psi_p(x_i, x_j)$ is to enforce consistency between the features of nearby pixels, encouraging smoothness in the prediction. To compute the pairwise potential, we utilize neural networks to learn a function that takes into account both the feature similarity between the nodes and their relative spatial positions. The pairwise potential is expressed as

$$\psi_p\left(x_i, x_j\right) = \text{Softmax}\left(q_i \cdot k_j^T + P\right) \cdot X \tag{12}$$

where q_i is the query vector associated with the feature map at pixel i. It is derived from the feature map F_{sar}^1 of pixel i and is learned by the neural network. k_j is the key vector

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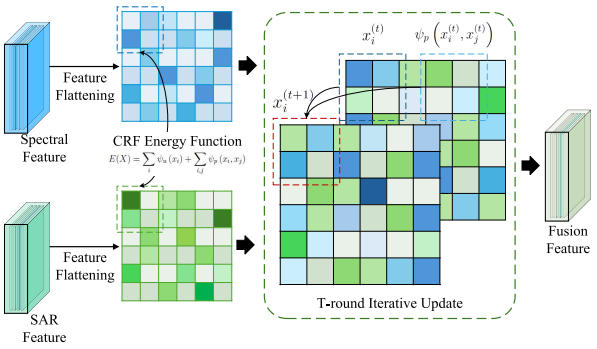


Fig. 4. Illustration of window-based CRF-inspired feature refinement.

associated with pixel j, similarly derived from the feature map $F^1_{\text{opt_hsi}}$ of pixel j. P is a relative position embedding, which incorporates spatial information to reflect the distance or relative positioning between pixels i and j. This helps capture the spatial relationships between neighboring pixels. X is the matrix of predicted values, which is weighted by the computed pairwise potential to refine the predictions.

To address the high computational cost of FC-CRFs over entire feature maps, we partition the feature maps into smaller $N \times N$ windows. The FC-CRF operation is then applied within each window independently, preserving local contextual dependencies while ensuring computational feasibility. This localized design significantly reduces the computational burden compared to global modeling. The complexity per window is expressed as

$$\Omega\left(\text{Window FC}\right) = hw \times \Omega\left(\psi_u\right) + hw \times \left(N^2 - 1\right) \times \Omega\left(\psi_p\right) \quad (13)$$

where hw is the number of windows, N^2 denotes the number of patches per window, and ψ_u, ψ_p represent unary and pairwise operations, respectively.

The iterative feature update mechanism is inspired by the mean-field approximation used in FC-CRFs [41], where each node refines its state by aggregating contextual information from its neighbors through pairwise interactions. Although our formulation does not explicitly model label distributions, it embodies a similar principle of message passing at the feature level to enhance local consistency. Specifically, we employ an update rule in which each node (i.e., feature vector) receives learned messages from its neighboring nodes within the same window. This is defined as

$$x_i^{(t+1)} = x_i^{(t)} + \eta \cdot \sum_{j \in \text{Window}} \psi_p\left(x_i^{(t)}, x_j^{(t)}\right) \quad (14)$$

where $x_i^{(t)}$ is the feature at iteration t, η is a step size parameter, and ψ_p serves as a learned pairwise potential modeling contextual affinity between features. This formulation enables structured and iterative refinement of feature representations by promoting semantic coherence within each local window. A visual depiction of this mechanism is provided in Fig. 4 to further clarify the spatial message propagation and update process.

To further address the locality constraint imposed by fixed window sizes, we introduce a global information aggregation module after each stage of feature refinement. Specifically, we apply multiscale global pooling to extract holistic context and fuse it with intra-window representations, thereby enhancing the model's ability to capture long-range dependencies across windows. This design ensures that spatial and spectral consistency is maintained both locally and globally, allowing the feature representation to progressively converge through iterative updates from higher to lower levels.

D. Class-Guided Contrastive Aggregation

Current contrastive learning methods primarily rely on fixed or single-scale feature matching, often neglecting the crucial requirement for consistency across multiple scales. This oversight limits their ability to effectively capture the intricate cross-scale relationships inherent in multimodal remote sensing data. To address the challenges posed by representation discrepancies among low-resolution hyperspectral images and high-resolution optical and SAR images, we propose a novel approach termed CG-MCA. This method seeks to enhance the fusion of data from varying resolutions and modalities while improving inter-class discrimination and intra-class consistency.

We extract multilevel features from various scales and modalities, capturing both global semantic information and local details. Let F^l_{fusion} denote the features extracted from the lth layer. A class-centered feature aggregation mechanism is employed to compute class centers, which represent the most representative features of each class at a specific scale. For a given class k, the class center $\mathbf{C}_k$ is defined as

$$\mathbf{C}_k = \frac{1}{N_k} \sum_{i \in \mathcal{P}_k} \sum_{l=1}^{4} F^l_{\text{fusion}} \quad (15)$$

where N_k is the number of pixels in class k, and $\mathcal{P}_k$ denotes the pixel indices belonging to class k. Class centers serve as reference points for contrastive learning, promoting intra-class compactness and maximizing inter-class separation by increasing the distance between class centers of different classes.

The contrastive loss encourages features of the same class to cluster around their respective class centers while maximizing the separation between different class centers. It is defined as

$$\mathcal{L} = \sum_k \left(\sum_{i,j \in \mathcal{P}_k} \mathcal{D}\left(\mathbf{f}_i, \mathbf{C}_k\right) + \sum_{k \neq j} \mathcal{D}\left(\mathbf{C}_k, \mathbf{C}_j\right) \right) \quad (16)$$

where $\mathcal{D}(\cdot,\cdot)$ cosine similarity metric, $\mathbf{f}_i$ represents the feature vector of feature point i. This formulation mitigates information loss and fusion errors caused by multiscale differences and modality misalignment. We further refine the contrastive loss by introducing intra-class compactness and inter-class separation terms

$$\mathcal{L}_{\text{contrast}} = \sum_k \mathcal{L}_{\text{intra}} + \lambda \sum_{k \neq j} \mathcal{L}_{\text{inter}} \quad (17)$$

where

$$\mathcal{L}_{\text{intra}} = \sum_{i,j \in \mathcal{P}_k} \left(1 - \cos\left(\mathbf{f}_i, \mathbf{C}_k\right)\right) \quad (18)$$

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$$\mathcal{L}_{\text{inter}} = \max\left(0, \epsilon - \cos\left(\mathbf{C}_k, \mathbf{C}_j\right)\right) \tag{19}$$

where λ is a hyperparameter balancing intra-class compactness and inter-class separation, and ϵ is a margin parameter that enforces inter-class separation.

To optimize the feature fusion process, the contrastive loss $\mathcal{L}_{\text{contrast}}$ is combined with the cross-entropy loss $\mathcal{L}_{\text{CE}}$ to form the overall loss function. While $\mathcal{L}_{\text{contrast}}$ enhances feature consistency and discrimination during fusion, $\mathcal{L}_{\text{CE}}$ ensures accurate semantic segmentation. The cross-entropy loss is commonly employed to measure the discrepancy between the predicted class distribution and the ground truth class distribution for each pixel. The cross-entropy loss can be mathematically expressed as follows:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log\left(p_{i,c}\right) \tag{20}$$

where N is the total number of pixels, C is the number of classes, $y_{i,c}$ is the ground truth label, and $p_{i,c}$ is the predicted probability for pixel i belonging to class c. The overall loss function is expressed as

$$\mathcal{L}_{\text{final}} = \mathcal{L}_{CE} + \lambda\mathcal{L}_{\text{contrast}} \tag{21}$$

where λ balances the contributions of semantic segmentation accuracy and feature consistency.

IV. Experiment

In this section, we perform extensive experiments on the two regions of our created dataset: Suzhou and Wuhan.

A. Datasets

This study utilizes multimodal satellite data, comprising optical, SAR, and hyperspectral imagery, collected from Suzhou region in Jiangsu Province and Wuhan region in Hubei Province, China. These datasets provide valuable insights into the land cover and environmental characteristics of the study areas, enabling a comprehensive analysis of their spatial and environmental diversity.

The optical data, sourced from the GF-2 satellite, offers high-resolution multispectral imagery with a spatial resolution of 3.2 m, capturing detailed surface features critical for land use and environmental monitoring. The SAR data, acquired from the GF-3 satellite, delivers all-weather, all-time imaging with a spatial resolution of 3 m, providing essential structural and textural information for land cover analysis in both Suzhou and Wuhan. Furthermore, hyperspectral data from the GF-5 satellite, with a spatial resolution of 20–40 m and spanning 330 spectral bands, allows for precise analysis of surface material composition and vegetation cover. This includes 150 bands in the visible and near-infrared (VNIR) regions and 180 bands in the short-wave infrared (SWIR) region, facilitating in-depth environmental assessments. Each modality offers unique and complementary information, contributing to a robust evaluation of land cover types and environmental conditions across these diverse regions. Although the modalities were acquired over a period of two to five months, all acquisitions were conducted within the same season (autumn to early winter). This setup reflects practical multimodal sensing scenarios while minimizing semantic shifts due to seasonal variations.

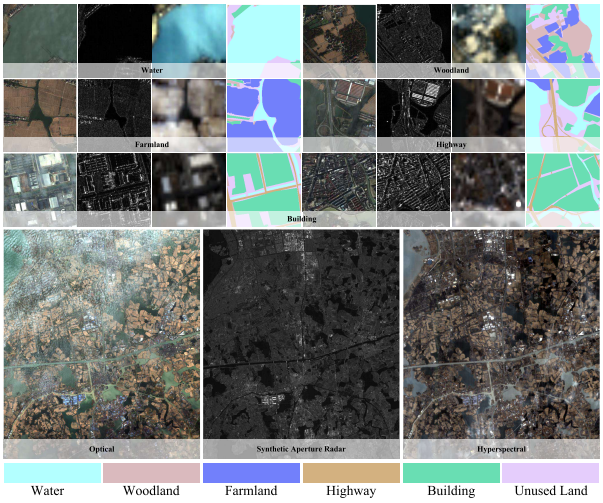


Fig. 5. Display of multimodal data in Suzhou area.

The dataset construction process is designed to retain the distinct resolution characteristics of each modality while ensuring spatial alignment. Optical and SAR images were first co-registered using georeferencing techniques to achieve pixel-level correspondence. Hyperspectral images, originally low in spatial resolution, were upsampled via bicubic interpolation for alignment with the optical imagery. After alignment, they were downsampled back to the original resolution and paired with the original hyperspectral images to preserve spectral integrity. This strategy maximizes the retention of modality-specific information while achieving geometric consistency. The resulting dataset enables rigorous evaluation of fusion techniques under realistic, resolution-diverse conditions.

1) Suzhou-Area: Suzhou, located in the southeastern part of Jiangsu Province, China, encompasses both urban and rural landscapes, offering a diverse terrain for analysis. The region is known for its intricate water networks, historical landmarks, and rapid urbanization. The dataset includes multimodal imagery—optical, SAR, and hyperspectral data—collected over an area of approximately 399.91 km^2, capturing a diverse set of land cover types. Temporally, Suzhou optical data were acquired in November 2023, SAR in July 2023, and hyperspectral (Gaofen-5) imagery in December 2023. Geographically, the study area is defined by the following coordinates: the northwest corner (120°34'23.96"E, 31°7'24.78"N), the northeast corner (120°45'41.18"E, 31°5'17.84"N), the southeast corner (120°42'46.66"E, 30°53'49.23"N), and the southwest corner (120°31'30.78"E, 30°55'56.04"N).

The land cover within the study area has been manually annotated into six categories: water, woodland, farmland, highway, building, and unused land. Examples of the annotated categories are shown in Fig. 5. The pixel distribution across these categories is as follows: water (19.26%), woodland (8.43%), farmland (23.52%), highway (2.63%), building (18.47%), and unused land (27.62%). The dataset consists of 621 image tiles, with the data split into training

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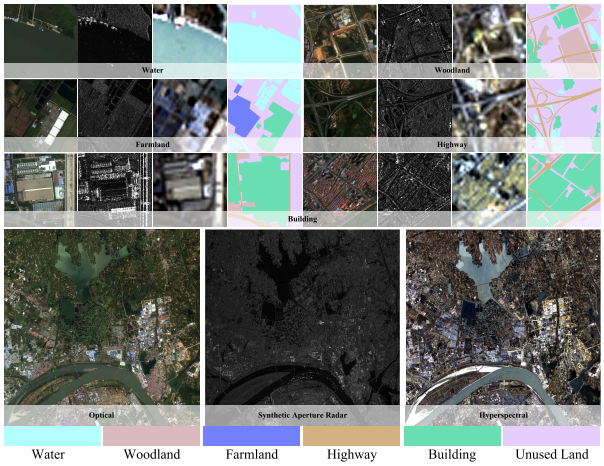


Fig. 6. Display of multimodal data in Wuhan area.

and evaluation sets at an 8:2 ratio, allocating 80% for training and 20% for evaluation.

2) Wuhan-Area: Wuhan dataset, collected from a central region of China, offers a comprehensive view of the diverse urban and industrial landscapes, providing a stark contrast to Suzhou region. This dataset includes multimodal imagery—optical, SAR, and hyperspectral data—capturing a range of land cover types typical of urban, industrial, and natural reserve areas. Optical data were acquired in September 2023, SAR in October 2023, and hyperspectral imagery in December 2023, with all modalities captured within the same season to reduce potential semantic variations. The geographic coordinates for Wuhan study area are as follows: northwest corner (114°24'0.71"E, 30°52'5.86"N), northeast corner (114°39'53.58"E, 30°49'33.70"N), southeast corner (114°36'52.96"E, 30°36'26.36"N), and southwest corner (114°21'2.26"E, 30°38'58.46"N), covering a total area of approximately 638.34 km^2.

Land cover within Wuhan study area has been annotated into six categories: water, woodland, farmland, highway, building, and unused land. Examples of the annotated categories are shown in Fig. 6. The pixel distribution across these categories is as follows: water (22.36%), woodland (4.31%), farmland (0.93%), highway (2.13%), building (15.48%), and unused land (54.78%). The dataset consists of 841 image tiles and is split into training and evaluation sets at an 8:2 ratio. Compared to Suzhou region, Wuhan dataset displays distinct land cover characteristics, with a higher proportion of unused land and lower percentages of farmland and woodland. These differences highlight the urban and industrial nature of Wuhan, contrasting with the more traditional water town landscape of Suzhou.

B. Experimental Setup

1) Implementation Details: Our model is implemented on the PyTorch framework. The VMamba [29] encoder is used as the backbone, with an embedding dimension of 512. We adopt the AdamW optimizer [42] with a weight decay of 0.01. The initial learning rate is set to 6×10^{-5}, and a polynomial learning rate schedule is employed throughout training. Regarding the input data, the optical images have a size of $256 \times 256 \times 3$, the SAR images are 256×256, and the hyperspectral images are $32 \times 32 \times 330$. The training process runs for 200 epochs on a single 4090 GPU, with a batch size of 8 for both Suzhou and Wuhan datasets.

2) Comparative Experimental Method: To comprehensively assess the efficacy of our proposed model, we conducted both visual comparison and quantitative analysis. In our evaluation, we benchmarked its performance against several leading multimodal semantic segmentation methods, including Deeplabv3+ [43], CMX [44], CMNeXt [45], Dformer [46], and Sigma [47]. Sigma is a multimodal semantic segmentation network that employs a Siamese architecture and Mamba fusion mechanism, achieving global receptive fields with linear complexity, thereby effectively enhancing performance in multimodal perception tasks. CMX is a framework for RGB-X semantic segmentation that employs cross-modal fusion with Transformers, integrating CNNs and Transformers for feature interaction and fusion. CMNeXt is a pioneering framework for arbitrary-modal semantic segmentation, employing a dual-branch Hub2Fuse paradigm to dynamically select informative features from multiple modalities and fuse them effectively, enhancing segmentation performance across various datasets. DFormer achieves outstanding performance and efficiency in RGB-D semantic segmentation by pretraining on ImageNet-1K to encode joint RGB-D representations and utilizing custom blocks for integrated RGB-depth feature encoding. DeepLabv3+ enhances semantic segmentation by refining object boundaries with a decoder module and employing depthwise separable convolutions for efficiency.

3) Evaluation Indicators: For the quantitative evaluation, we have used several widely used metrics, such as the class accuracy (CA), the overall accuracy (OA), the mean Intersection over Union (mIoU), and the frequency-weighted IoU (Freq IoU). These metrics provide a comprehensive evaluation of the segmentation performance of our model. Higher values for these metrics generally indicate better classification accuracy in remote sensing image classification tasks.

C. Experimental Results on Suzhou Area

Suzhou area data consists of 497 training images and 124 test images, covering a variety of land cover classes such as water, woodland, farmland, highway, building, and unused Land, with a total of six classes, and some of the visualized images are shown in Fig. 5. Our experimental setup is described above. Table I presents a comparative analysis of the performance of the various algorithms on Suzhou area data. The last row of Table I highlights the performance of our proposed SGIFN network. On Suzhou area data, this network has the highest OA, mIOU, and Freq IoU compared to other methods. In addition, Fig. 7 shows some of the visualized experimental results of our proposed method and other comparative methods.

Table I presents the performance comparison of different methods on Suzhou dataset, evaluated using three key metrics: mIoU, Freq IoU, and OA. As shown in the table, our method achieves the highest score of 69.61% for mIoU,

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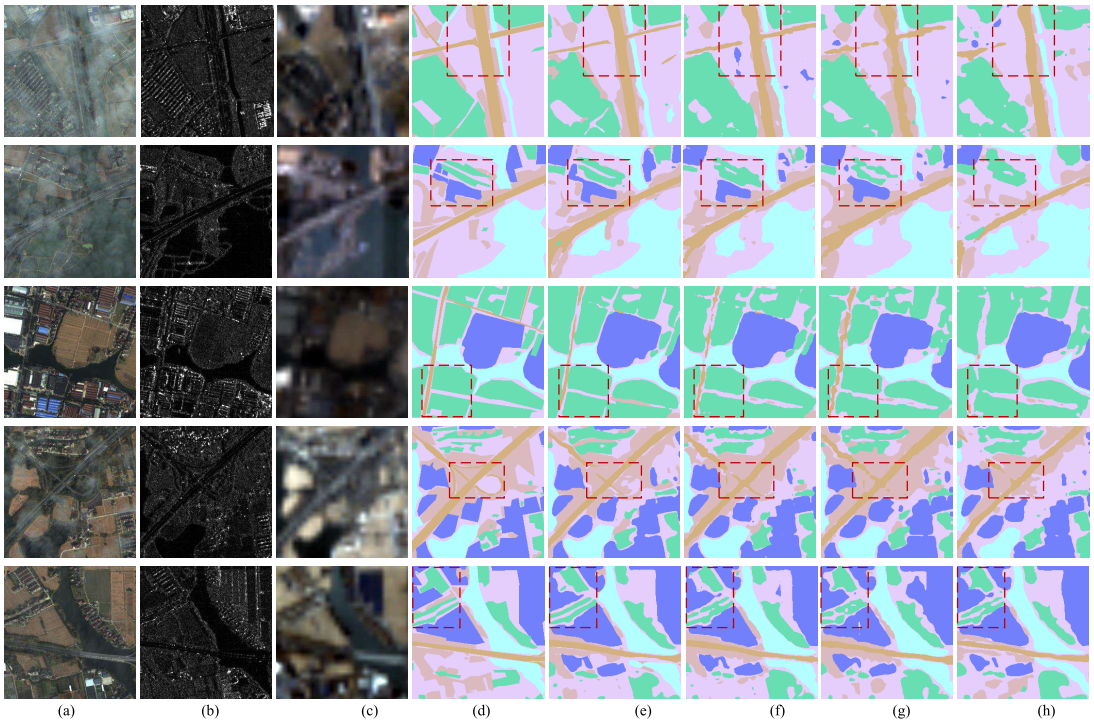


Fig. 7. Results of semantic segmentation obtained using different approaches for Suzhou area data. (a) Optical images.(b) SAR images. (c) Hyperspectral images. (d) Ground truth image. (e) Ours. (f) CMX. (g) DFormer. (h) Sigma.

TABLE I
COMPARISON OF THE RESULTS OF DIFFERENT MULTIMODAL FUSION METHODS ON SUZHOU AREA DATA

Methods	Water	Woodland	Farmland	Highway	Building	Unused Land	mIoU(%)	Freq IoU(%)	OA(%)
Deeplabv3+ [43]	87.30	32.21	84.18	34.68	69.51	49.42	59.55	67.67	79.36
CMX [44]	90.89	**46.68**	**86.45**	53.88	76.50	57.89	68.71	74.11	84.07
CMNeXt [45]	90.30	44.59	86.07	49.74	76.69	57.12	67.42	73.44	83.79
DFormer [46]	89.68	46.13	85.75	46.87	75.53	55.89	66.64	72.77	83.33
Sigma [47]	88.97	36.94	84.92	46.61	72.95	52.85	63.87	70.40	81.46
Ours	**91.49**	46.43	86.16	**57.73**	**77.71**	**58.14**	**69.61**	**74.52**	**84.49**

reflecting the best overall segmentation performance among all methods. Regarding Freq IoU, our approach reaches 74.52%, surpassing the other methods and indicating its robustness in handling class imbalances. In addition, in terms of OA, our model achieves 84.49%, further highlighting its excellence in overall classification accuracy. When analyzing individual land cover categories, our method consistently ranks among the top performers. For example, it achieves a remarkable 91.49% accuracy in the "Water" category and 77.71% in the "Building" category, both of which outperform competing methods. Although CMX and CMNeXt demonstrate strong results in some categories, their performance lacks consistency, particularly in more challenging classes such as "Highway" and "Unused Land." In contrast, our proposed method achieves balanced improvements across all categories, with notable advances in "Highway" (57.73%) and "Unused Land" (58.14%). These results underscore the advantages of our method in effectively addressing class imbalances and achieving comprehensive improvements across diverse land cover categories. The superior performance across mIoU, Freq IoU, and OA metrics further validates the robustness and effectiveness of our approach in multimodal data fusion and semantic segmentation tasks.

Fig. 7 illustrates the semantic segmentation results obtained using different methods. Where (a) is an optical image, (b) is a SAR image, (c) is a hyperspectral image (HSI), and (d) is a real labeled ground truth image, while (e)–(h) show the segmentation results of our proposed method as well as of the CMX, Dformer, and Sigma methods, respectively. As shown in the figure, the proposed method demonstrates improved structural completeness and higher classification accuracy, reflected by clearer object boundaries and reduced misclassification.

D. Experimental Results on Wuhan Area

Wuhan area data includes 673 training images and 168 testing images, encompassing six distinct land cover categories: water, woodland, farmland, highway, building, and unused land. A selection of visualized samples from the dataset is displayed in Fig. 6. The experimental setup for the evaluation has been described previously. Table II provides

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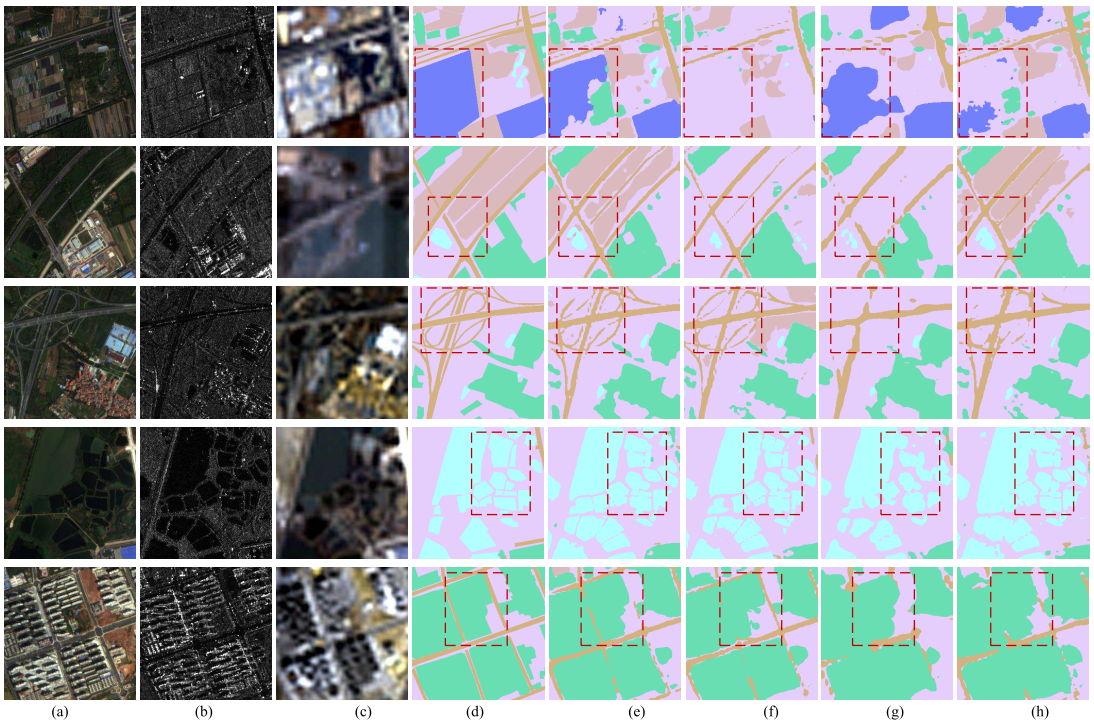


Fig. 8. Results of semantic segmentation obtained using different approaches for Wuhan area data. (a) Optical images.(b) SAR images. (c) Hyperspectral images. (d) Ground truth image. (e) Ours. (f) CMX. (g) DFormer. (h) Sigma.

TABLE II
COMPARISON OF THE RESULTS OF DIFFERENT MULTIMODAL FUSION METHODS ON WUHAN AREA DATA

Methods	Water	Woodland	Farmland	Highway	Building	Unused Land	mIoU(%)	Freq IoU(%)	OA(%)
Deeplabv3+ [43]	61.6	4.82	23.22	48.48	74.44	73.36	47.65	66.77	80.62
CMX [44]	79.75	30.77	12.07	58.95	77.04	78.91	56.25	75.70	86.03
CMNeXt [45]	82.29	28.91	41.38	**63.86**	**77.58**	79.59	62.27	77.07	86.92
DFormer [46]	68.91	16.00	21.99	39.84	71.07	74.54	48.73	68.96	81.58
Sigma [47]	80.05	34.40	27.69	56.04	75.57	77.79	58.59	75.20	85.72
Ours	**83.27**	**38.82**	**42.26**	62.50	76.40	**80.48**	**63.96**	**78.03**	**87.54**

a detailed performance comparison of various algorithms on this dataset. Notably, the final row in the table highlights the results achieved by our proposed SGIFN network, which outperforms other methods in terms of OA, mIoU, and Freq IoU. In addition, Fig. 8 illustrates visual comparisons of the segmentation results produced by our approach and the baseline methods, further demonstrating the superiority of our model.

Table II summarizes the performance comparison of various methods on Wuhan dataset, evaluated using three primary metrics: mIoU, Freq IoU, and OA. Our method achieves the highest mIoU of 63.96%, demonstrating superior overall segmentation capability compared to all competing approaches. For Freq IoU, our method records 78.03%, outperforming others and highlighting its ability to effectively address class imbalance. Furthermore, with an OA of 87.54%, our approach showcases its outstanding accuracy in overall classification tasks. Notably, it achieves an impressive accuracy of 83.27% in the "Water" category and 38.83% in the "Woodland" category, surpassing the results of competing methods. While CMNeXt exhibits strong performance in certain categories, it demonstrates inconsistencies, especially in more challenging classes like "Woodland" and "Farmland." In contrast, our method delivers balanced improvements across all classes, achieving significant gains in "Woodland" (38.83%) and "Farmland" (42.26%). These results emphasize the effectiveness of our method in mitigating class imbalance and delivering comprehensive performance enhancements across diverse land cover categories. The consistent superiority of our method across mIoU, Freq IoU, and OA metrics further affirms its robustness and efficacy in multimodal data fusion and semantic segmentation.

Fig. 8 provides a visual comparison of the segmentation results obtained by different methods. Fig. 8(a) displays the optical image, (b) shows the SAR image, (c) depicts the hyperspectral image (HSI), and (d) represents the ground-truth labeled image. Fig. 8(e)–(h) showcase the segmentation results of our proposed method alongside those of CMX, DFormer, and Sigma, respectively. The visualizations clearly illustrate that our method produces more structurally

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TABLE III
SEGMENTATION PERFORMANCE OF OUR METHOD WITH DIFFERENT MODAL COMBINATIONS

Data	mIoU(%)	Freq IoU(%)	OA(%)
Optical & HSI & SAR	69.61	74.52	84.49
Optical & SAR	66.44	72.38	82.86
Optical & HSI	62.70	69.61	80.89
HSI & SAR	58.00	65.75	78.16
Optical	62.17	67.92	79.64
SAR	41.94	49.72	63.18
HSI	52.54	62.40	75.92

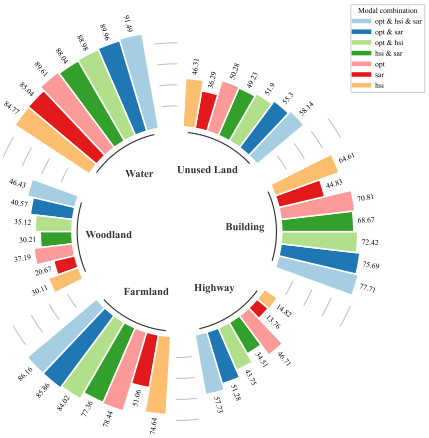


Fig. 9. Grouped circular of the segmentation performance across different land cover categories for modal combinations.

complete and accurate classification results, further underscoring its advantages in this task.

E. Effectiveness of Different Modalities

We investigated the effect of various modality combinations on the semantic segmentation task, as summarized in Table III. The results indicate that integrating all three modalities—optical, hyperspectral imaging (HSI), and SAR—yields the highest segmentation performance. This demonstrates that combining complementary information from these three modalities significantly enhances segmentation accuracy. Among the bimodal combinations, Optical & SAR outperformed the others, achieving an mIoU of 66.44%. This suggests that the combination of SAR and Optical data provides complementary features that are highly beneficial for segmentation. Optical data alone achieved the highest mIoU (62.17%), thanks to its high-resolution spatial information, while SAR data alone resulted in the lowest performance, with an mIoU of 41.94%, due to its limited spectral resolution and noisy nature.

To further illustrate the performance across different land cover categories, grouped circular charts are presented in Fig. 9. These charts highlight the segmentation quality for different categories, including water, woodland, farmland, highway, building, and unused land. While single-modality SAR images lack spectral information, they perform relatively well for the water category due to the distinct scattering characteristics of SAR, which effectively differentiate water bodies from other land cover types. Consequently, the SAR modality does not significantly lag behind other multimodal combinations in the water body category. However, the performance of SAR alone is much lower in other categories, mainly because these categories depend heavily on spectral information, which SAR cannot provide. When fused with optical images, segmentation performance improves notably. For example, in the Highway category, the IoU increases to 51.28%, representing a 37.52% improvement over single-modality SAR and an additional 4.57% improvement over single-modality Optical images. This highlights the unique contribution of SAR in enhancing segmentation performance, particularly in geometry-dominated categories like highways. The texture and geometric features provided by SAR images are critical in improving segmentation accuracy when integrated with other modalities.

TABLE IV
IMPACT OF SSID AND SGS-CRFS ON SEGMENTATION PERFORMANCE

SSID	SGS-CRFs	mIoU(%)	Freq IoU(%)	OA(%)
✓	✓	69.61	74.52	84.49
✓		66.82	71.97	82.70
	✓	65.43	71.78	82.47
		67.17	70.24	81.69

F. Ablation Study on the Impact of Multimodal Fusion Components

In this section, we conduct an ablation study to evaluate the contributions of two essential components in our proposed method: the SSID and SGS-CRFs. The experimental results are presented in Table IV.

When both SSID and SGS-CRFs are incorporated together, the model achieves the highest performance across all evaluation metrics, with a mIoU of 69.61%, Freq IoU of 74.52%, and OA of 84.49%. This demonstrates the complementary nature of SSID and SGS-CRFs: SSID effectively facilitates the integration of spectral features from diverse modalities, such as optical and hyperspectral images, which enhances the model's ability to leverage rich spectral information. Meanwhile, SGS-CRFs refine segmentation boundaries and reduce noise, improving segmentation accuracy. The combined application of these two components results in a significant performance boost, indicating their synergistic effect in addressing the challenges inherent in multimodal semantic segmentation. In contrast, when only SSID is employed (without SGS-CRFs), the model's performance slightly decreases, with mIoU, Freq IoU, and OA values of 66.82%, 71.97%, and 82.70%, respectively. While SSID successfully integrates spectral information from multiple modalities, the absence of SGS-CRFs limits the refinement of segmentation boundaries, leading to a reduction in segmentation accuracy. Similarly, when only SGS-CRFs are used (without SSID), the performance further declines, with mIoU, Freq IoU, and OA values of 65.43%, 71.78%, and 82.47%, respectively.

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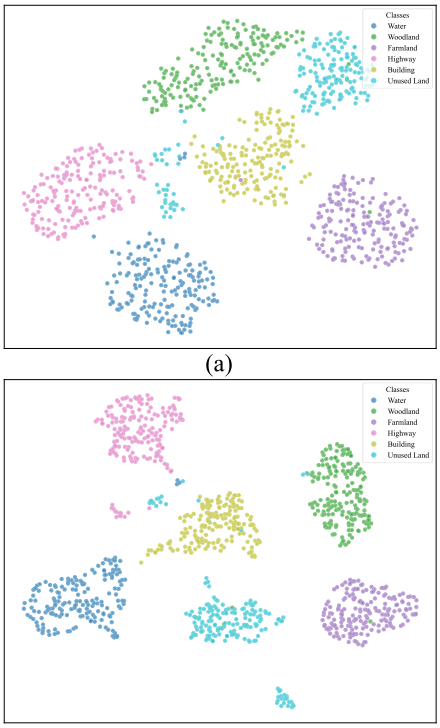


Fig. 10. UMAP visualization of feature distributions with and without CGCA. (a) Visualization on model without CGCA. (b) Visualization on model with CGCA.

The results of the ablation study highlight the critical roles of both SSID and SGS-CRFs in enhancing the performance of the proposed multimodal segmentation method. The individual contributions of each component result in a modest decrease in performance, underscoring the importance of their combined application in effectively addressing the complexities of multimodal semantic segmentation.

G. Ablation Study on the Impact of Class-Guided Contrastive Aggregation

This section presents an ablation study to investigate the effect of incorporating class-guided contrastive aggregation (CGCA) into our multimodal fusion-based semantic segmentation model. We compare the performance of the model with and without CGCA using both quantitative evaluation metrics and qualitative feature analysis via uniform manifold approximation and projection (UMAP) [48] visualizations.

The experimental results demonstrate that the inclusion of CGCA significantly improves the model's performance across all evaluated metrics. Specifically, the model with CGCA achieves a mIoU of 69.61%, a Freq IoU of 74.52%, and an OA of 84.49%. In contrast, the baseline model without CGCA reports a mIoU of 66.60%, Freq IoU of 72.32%, and OA of 82.85%. These improvements suggest that CGCA contributes to enhanced class separation and feature discrimination, which directly boosts the segmentation accuracy.

In order to further investigate the impact of CGCA on feature learning, we utilize UMAP to visualize the feature distribution in a 2-D space. Fig. 10 presents UMAP projections of the feature embeddings from both models (with and without CGCA). The UMAP technique provides insights into how well the model separates features from different classes in the high-dimensional space. Without CGCA, feature distributions exhibit significant overlap between classes, indicating poor separation and weak intra-class consistency. This overlap results in indistinct inter-class boundaries, which hinders the model's ability to effectively distinguish between similar classes, particularly in complex multimodal scenarios such as remote sensing data. On the other hand, with the integration of CGCA, the feature distribution shows clear separation between different classes, with features from the same class grouped more tightly and inter-class distances becoming more distinct. This improved clustering of features enhances the model's ability to distinguish challenging class pairs, such as buildings versus unused land, facilitating better segmentation accuracy. The enhanced feature representation enabled by CGCA thus plays a critical role in improving the overall performance of the segmentation model, particularly in scenarios involving complex, multimodal data.

H. Discussion

To address spatial and spectral inconsistencies arising from resolution and modality differences in multimodal semantic segmentation, this study proposes SGIFN, a unified framework. SGIFN achieves strong results on real-world datasets by integrating resolution-aware modules—SSID, SGS-CRF, and CG-MCA—while there is still potential for improvement through more advanced fusion and modeling techniques. For example, token-wise interaction mechanisms [49] could improve semantic alignment across modalities, especially during early feature disentanglement in SSID. In addition, combining graph neural networks (GNNs) with latent feature modeling [50] offers an effective way to capture spatial, spectral, and semantic dependencies across varying resolutions. These methods could enhance or replace parts of SGS-CRF, enabling more flexible message passing and richer fused representations.

Beyond architectural improvements, increasing data efficiency is important for applying SGIFN in broader remote sensing scenarios. Although this study uses newly annotated datasets from Wuhan and Suzhou, the lack of large-scale labeled data remains a challenge. To address this, remote sensing–specific data augmentation [51], especially for few-shot or imbalanced classes, can help improve generalization. Moreover, self-paced semi-supervised learning [52] can reduce reliance on labeled data by gradually incorporating high-confidence unlabeled samples during training. These strategies together can enhance SGIFN's robustness, scalability, and practical use in multimodal semantic segmentation.

V. Conclusion

We propose an SGIFN for the semantic segmentation of multimodal remote sensing images under resolution discrepancies. The SGIFN effectively integrates optical, hyperspectral, and SAR data to address the challenges posed by resolution

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differences between modalities. The SSID module refines cross-modal integration by iteratively aligning spectral and spatial features, enhancing hyperspectral-optical fusion. The SGS-CRF captures multimodal spectral-geometric dependencies with an FC-CRF and efficient window-based optimization, improving fusion accuracy and computational efficiency. In addition, the CG-MCA module improves feature consistency and separability across scales, ensuring robust segmentation results. Experimental evaluations on multimodal datasets demonstrate that SGIFN significantly outperforms existing methods in terms of segmentation accuracy and fusion quality.

It is difficult to acquire large-scale multimodal datasets with consistent resolution. Therefore, future work will focus on exploring the use of synthetic data and semi-supervised learning techniques to further enhance the performance of multimodal segmentation models.

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